

AML Fraud Detection Report
Integrated AML & Transaction Risk Management Platform

Total transactions scanned: 1,000
Flagged for the watchlist: 600 (60.0%)
 Requiring analyst review: 146 (Critical + High severity)
 Critical severity: 14
 High severity: 132
 Medium severity (monitor only): 454

Detection rules applied:

Behavioural score (0-4, one point each):

R1 Large single transaction (> \$9,000)

R2 High transaction frequency for the customer (>= 4 txns)

R3 High-risk transaction location (source is_high_risk flag)

R4 Transacting bank compliance score < 0.6

Sanctions flag (0-1):

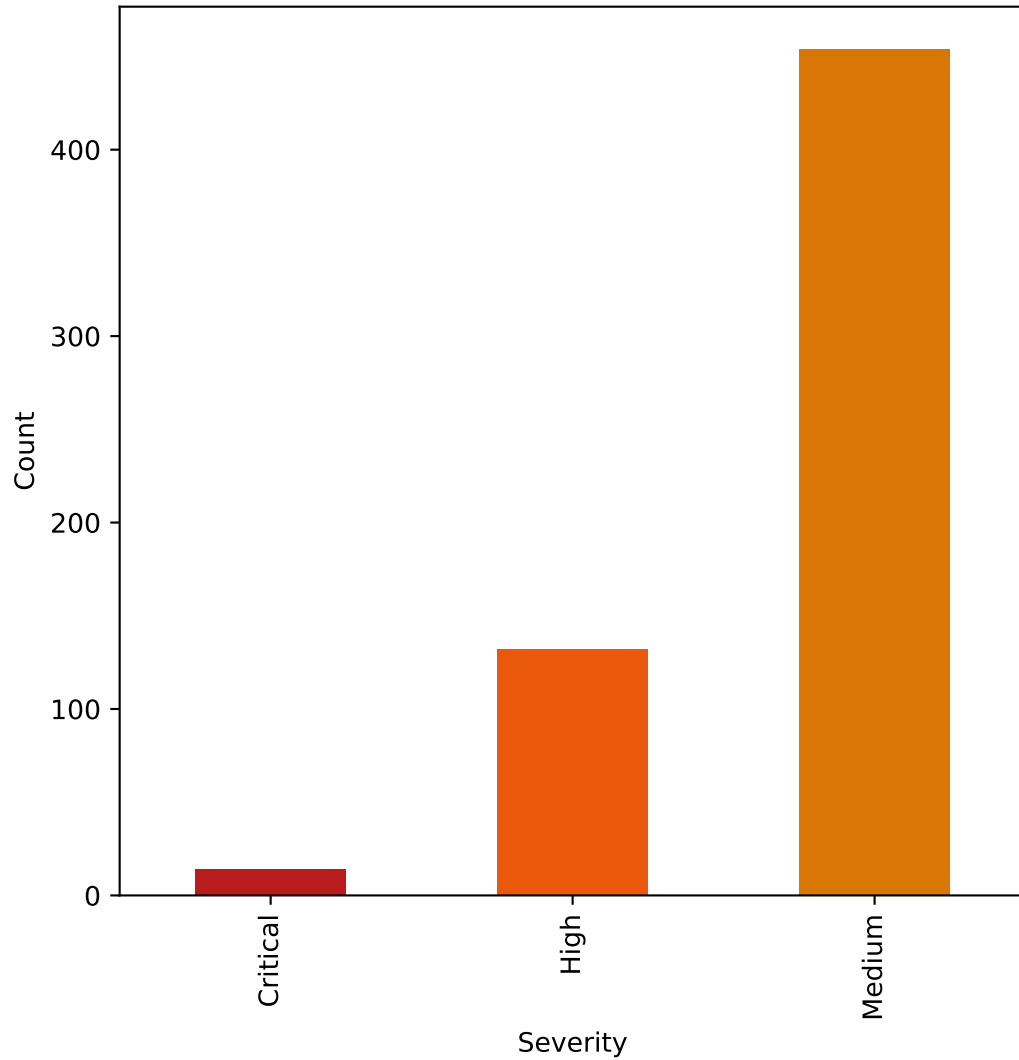
R5 Customer or transacting bank has sanctions-list history

fraud_score = behavioural score + sanctions flag (range 0-5)

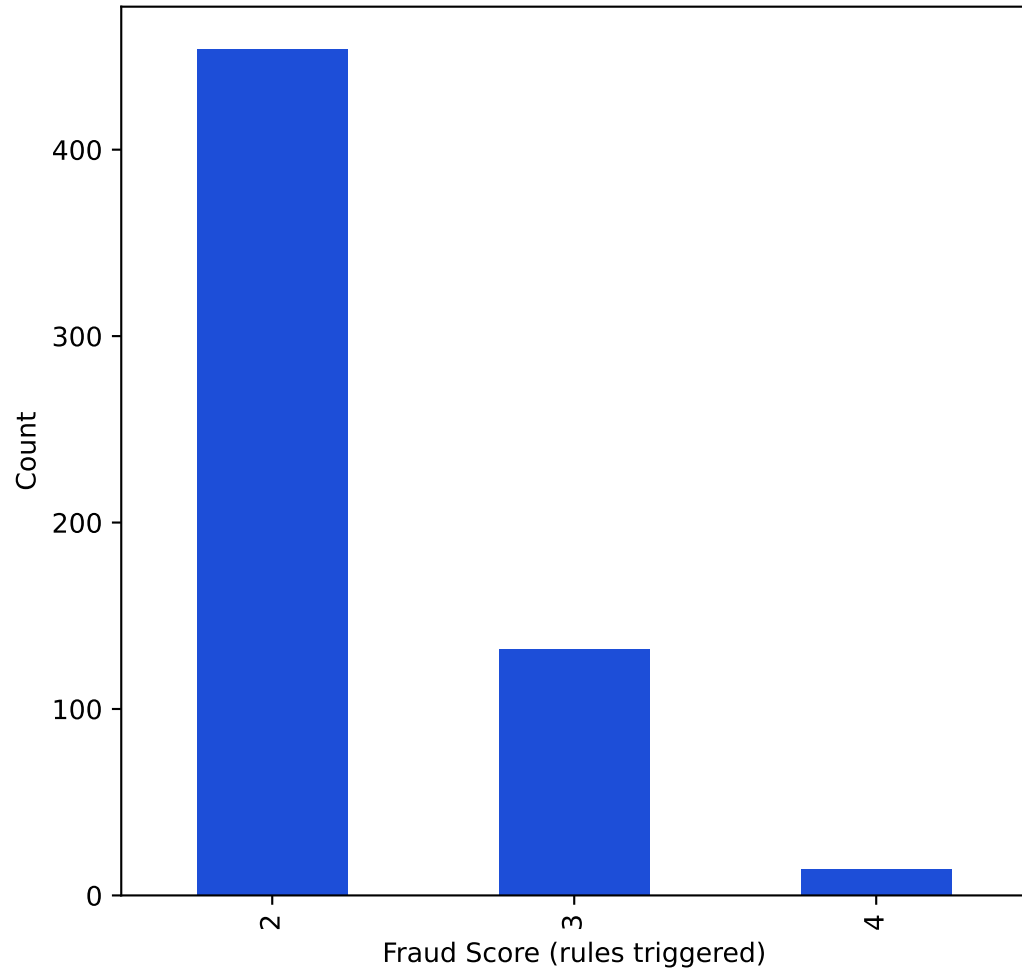
Flag threshold: fraud_score >= 2. Review threshold: fraud_score >= 3.
(Sanctions coverage in this dataset is unusually broad -- ~85% of transactions touch a sanctioned party -- so it is deliberately weighted lower than the behavioural signals; see sql/04_fraud_detection_rules.sql for the full rationale.)

Note: False Positive Rate and Detection Accuracy require analyst disposition data not present in the source files. A disposition log template (reports/fraud_disposition_log_template.csv) is included so these KPIs can be tracked once reviews are logged.

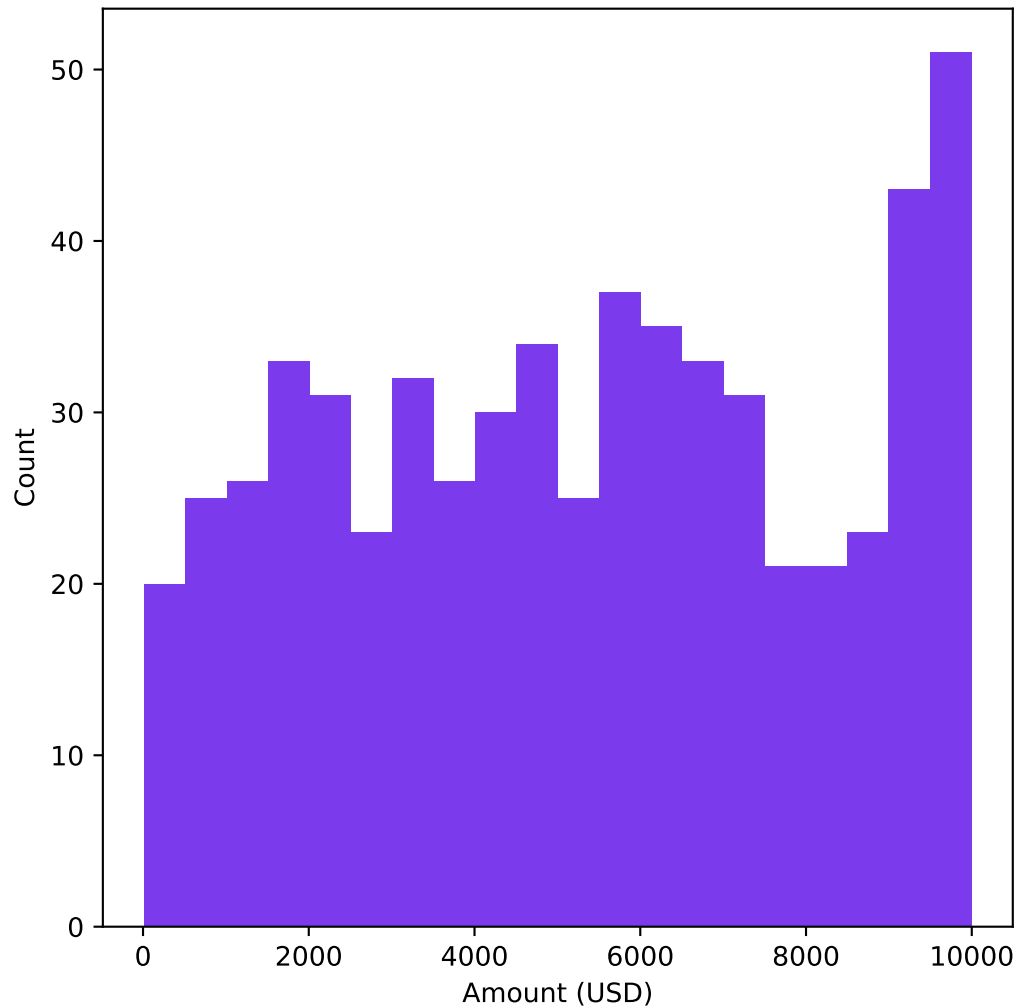
Flagged Transactions by Severity



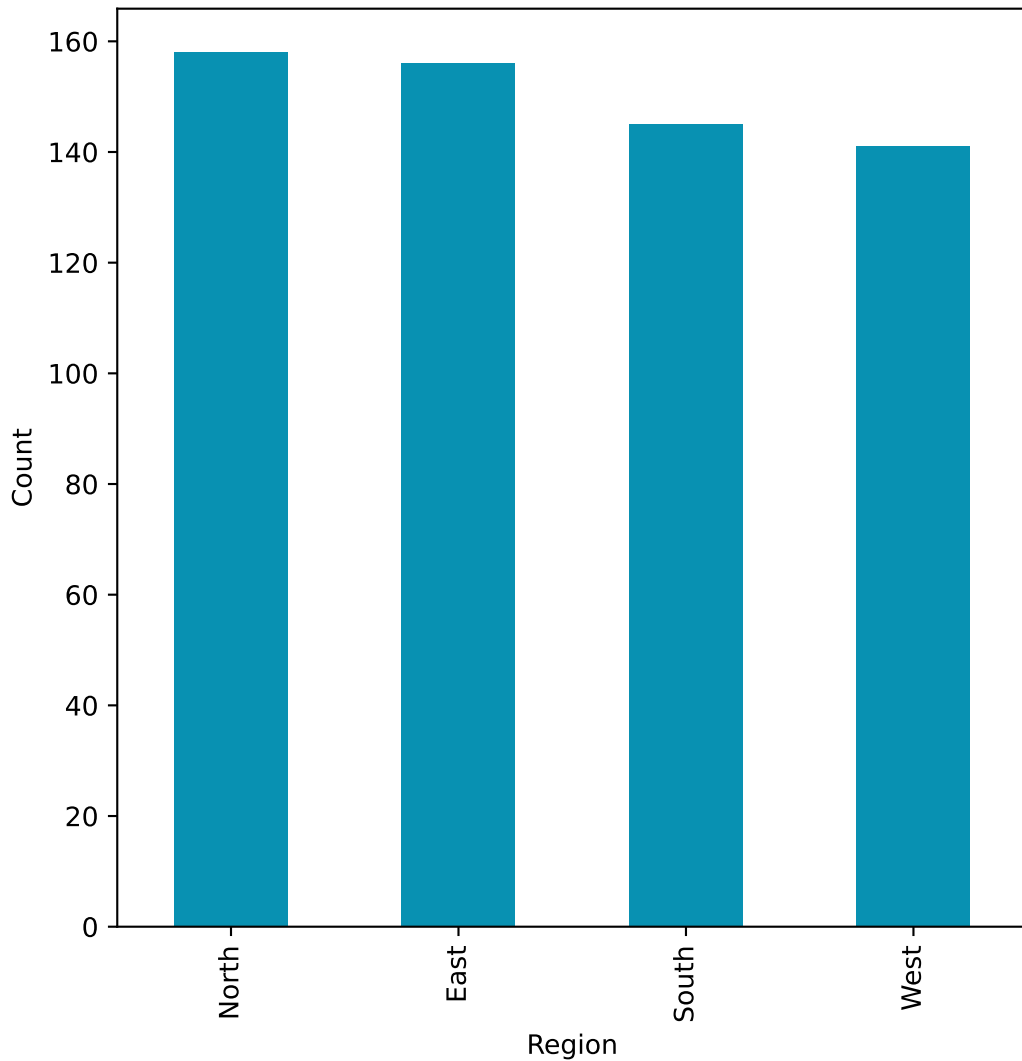
Flagged Transactions by Fraud Score



Flagged Transaction Amount Distribution



Flagged Transactions by Customer Region



Top 20 Highest-Risk Flagged Transactions

transaction_id	customer_name	bank_name	amount	fraud_score	severity
501	Marcus Little	Bank 809	9994.12	4	Critical
566	Daniel Mason	Bank 147	9934.56	4	Critical
639	Timothy Contreras	Bank 535	9790.33	4	Critical
35	Deborah Reyes	Bank 958	9716.82	4	Critical
954	Lisa Colon	Bank 557	9683.82	4	Critical
909	Jennifer Jimenez	Bank 132	9648.45	4	Critical
567	Cory Collins	Bank 88	9509.48	4	Critical
17	Anna Patterson	Bank 759	9417.45	4	Critical
667	Angel Harper	Bank 535	9387.23	4	Critical
323	Matthew Patel	Bank 514	9355.9	4	Critical
3	Mark Taylor	Bank 598	9223.78	4	Critical
330	Alexis Wilson	Bank 651	5910.77	4	Critical
238	Sharon Crawford	Bank 677	4561.89	4	Critical
214	Mary Brewer	Bank 746	4037.01	4	Critical
178	Kathryn Shelton	Bank 898	9995.9	3	High
380	Douglas Martin	Bank 48	9931.19	3	High
900	Brenda Johnson	Bank 122	9928.75	3	High
500	Valerie Robinson	Bank 732	9895.62	3	High
188	Haley Harris	Wilcox, Lynch and Weeks	9855.67	3	High
571	Nathan Green	Bank 254	9851.24	3	High